

ARITRA MUKHOPADHYAY

Kolkata, West Bengal, India | 8100060437 | mukhoari@gmail.com | [Linkedin](#)

SUMMARY

B.Tech Mechanical Engineering student (KGEC, Batch 2027) with hands-on experience in CAD modelling, FEA simulation, CNC machining, and automotive fabrication. Convenor at the Students' Automobile Club of KGEC; led fabrication of a CV-category go-kart raced at the Indian Karting Race 2025. Trained in defence manufacturing (Ordnance Factory Dum Dum) and steel processing (SAIL Durgapur). Proficient in SolidWorks, AutoCAD, Ansys, Python automation, and graphic design.

EDUCATION

Kalyani Government Engineering College (KGEC) — B.Tech, Mechanical Engineering

Aug 2023 – Jun 2027

Kendriya Vidyalaya — Senior Secondary

2023

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, AutoCAD (2D/3D), Ansys Mechanical (Static Structural & Thermal FEA), Siemens NX

Programming & Automation: Python, GUI Development, Parametric Modelling, Arduino, U G-Code Sensor, iCNC Simulator

Manufacturing: CNC Turning & Milling, Mini-CNC Automation, Heat Treatment, Materials Testing (Hardness, Impact, Tensile)

Design: Adobe Illustrator, Adobe Photoshop, Adobe Lightroom, Canva, Figma, Graphic Design

PROJECTS

Automated CAD Engine — Python-Powered Parametric Spur Gear Generator

Aug 2026

- Built a desktop automation app integrating Python with the SolidWorks API to auto-generate 3D models and 2D drafts of spur gears from user-input parameters, cutting design-to-documentation time from minutes to under 5 seconds per iteration.

Finite Element Analysis of IC Engine Piston & Connecting Rod Assembly using ANSYS

Sept 2026

- Performed Static Structural and Steady-State Thermal FEA on a piston-connecting rod assembly, using path linearization along the rod shank and piston cross-section to extract quantitative deformation and thermal gradients.

Reducing Thermal Damage & Thrust Force in Orthopaedic Drilling — Research Project

Aug 2025 – Present

- Designed a low-cost, Arduino-controlled mini-CNC drilling setup and ran Taguchi L9 experiments on bone specimens with thermocouples and a load-cell dynamometer; built Minitab RSM models to jointly minimize temperature rise and thrust force.
- Awarded Best Paper Presentation at FOSET's 16th Academic Meet, 2026.

TAARZAN Go-Kart — Students' Automobile Club of KGEC

Oct – Nov 2025

- Part of a 23-member team that designed, fabricated, and raced a CV-category go-kart at the Indian Karting Race (IKR), Coimbatore, clearing technical inspection among 39 teams and securing a ₹5,000 cash prize.

INDUSTRIAL TRAINING

Vocational Trainee — Alloy Steels Plant, Steel Authority of India Limited (SAIL), Durgapur

Aug 2026

- Gained exposure to steel melting, secondary refining, continuous casting, rolling, forging, and heat treatment; studied the vacuum steam ejector degassing system and hydraulic furnace-control system.

Vocational Trainee — Ordnance Factory Dum Dum (OFDC), Ministry of Defence

Jul 2025

- Trained in CNC turning/milling, jig boring, laser cutting, and grinding; exposed to missile/bomb-fuse assembly, polymer component manufacturing, and Siemens NX product design. Completed a mini-project on end-to-end manufacturing of LRCCR/MCCR components (Nozzle Block, Aluminium Body).

POSITIONS OF RESPONSIBILITY

- Convenor, Students' Automobile Club of KGEC (Jul 2026 – Present; previously Associate and Member, May 2024 – Jul 2026)
- Secretary, IIW-India KGEC Students' Chapter (Jun 2026 – Present).
- Convenor, Shutterbug-KGEC (Photography Club) and Imposter-KGEC (Design Club), KGEC.
- Training & Placement Coordinator, Training and Placement Cell, KGEC (Jul 2025 – Present).

HONORS & AWARDS

- Best Paper Presentation — Forum of Scientists, Engineers & Technologists (FOSET), 16th Academic Meet, 2026.
- Best Presenter Award — 7-day Hands-on Workshop on Data-Driven Quality Control in Smart Manufacturing Systems, IIT (ISM) Dhanbad, 2026.

CERTIFICATIONS

The Complete Python Bootcamp: From Zero to Hero | Learn AutoCAD 2D & 3D: From Zero to Hero | SolidWorks Academy: Fundamental Course for Beginners | Siemens Mobility – Project Manager Job Simulation | Introduction to Soft Skills

EXTRACURRICULAR ACTIVITIES

- Graphic Designer — Promotional and Event material for ESPEKTRO the Annual Techno-Management Fiesta of KGEC and client work including for Avalanche.